SYSTEM ANALYSIS DOCUMENTATION (SAD)

Project Zync

The Sovereign AI for Rare Earth Engineering

UMHackathon 2026 Domain:

AI for Economic Empowerment & Decision Intelligence

Team: Gunners

Name	Matric Number
Lau Hiap Meng	23111208
Loh Yong Qiao	S2184898
Tai Jin Wei	23080642
Kan Penny	24004556
Valencian Seow Yun Sun	24004618

Email: umhackathon@um.edu.my

1. Introduction

Zync is the strategic technological solution designed to address Malaysia's engineering intelligence gap in the Rare Earth Element (REE) sector. While Malaysia possesses RM 1 trillion in mineral reserves, it lacks the domestic expertise to process them independently. Zync integrates **Z.ai's GLM-5.1** to act as a sovereign reasoning layer, augmenting local engineers into midstream plant architects capable of making expert-level hydrometallurgical decisions.

1.1. Purpose

This System Analysis Document (SAD) highlights the technical scope, architectural design and logic-driven decisions behind the Zync platform. It serves as the primary technical reference for developers, testers, researchers and stakeholders to understand the integration of agentic AI with industrial mineral processing workflows.

Key System Elements Covered:

- **Architecture:** A multi-agent, cloud-native ecosystem centered on the Z.ai GLM-5.1 reasoning engine.
- **Data Flows:** The transition of unstructured geological dark data and structured site logs into validated industrial blueprints.
- **Model Process:** An end-to-end autonomous pipeline involving six specialized AI agents for diagnosis, optimization and compliance.
- **Role of Reference:** This document ensures high-level alignment across the development lifecycle, focusing on scalability and sovereign data residency.

1.2. Background

Malaysia is committed to processing 16.1 million tonnes of REE domestically under the **13th Malaysia Plan (13MP)**. However, the reliance on foreign technological monopolies, particularly in advanced separation science remains a critical bottleneck. Zync is designed to provide the engineering depth required to solve complex hydrometallurgical problems in minutes rather than months.

Major Architectural Shift: Unlike traditional process simulation tools like METSIM which require expert human inputs to simulate known processes, Zync uses Agentic Engineering to reason through unknown process recipes autonomously. This represents a shift from static automation to Decision Intelligence.

1.3. Target Stakeholders

Stakeholder	Roles	Expectations
Site Engineers	Input weekly logs and execute corrective actions.	Immediate, expert-level diagnosis of yield failures.
Process Managers	Select leaching chemistry and plan extraction batches.	Data-driven lixiviant recommendations with ESG trade-off analysis.
Compliance Officers	Verify adherence to DOE and AELB regulations.	Real-time blocking of non-compliant configurations.
R&D Officers	Design pilot programs and prioritize extraction zones.	Strategic ranking of mineral zones based on economic and strategic yield.
Development Team	Build and maintain the agentic AI pipeline.	Modular codebase, clear API contracts and robust prompt versioning.

2. System Architecture & Design

2.1. High Level Architecture

2.1.1. Overview

Type	Details
System	Sovereign Command Center Web Dashboard
Architecture	Multi-Agent Microservices

Zync is primarily structured as a client-server-based system utilizing three distinct operational modules. Site engineers and process managers interact with the backend API layer to trigger specialized agentic workflows. Independent microservices manage diagnostic loops and strategic optimizations while sharing a centralized PostgreSQL database for operational persistence.

2.1.2 LLM as Service Layer

The architecture integrates GLM as a core service layer rather than a generic AI module. This allows the system to perform high-stakes scientific reasoning and multi-variable optimization across the entire processing pipeline.

2.1.3 Dependency Diagram

The REE-IES dependency architecture is organised across five distinct layers, each with a clearly defined responsibility boundary. Data flows vertically from the presentation layer down to persistent storage, passing through API validation, agent orchestration and sovereign inference at each stage.

Presentation Layer — React + Vite Frontend

The topmost layer is the React + Vite single-page console UI. Operators interact with three decision modules, Lixiviant Optimisation, Process Diagnosis and Zone Strategy, each constructing role-isolated prompts that combine mineralogical logic with real-time site data submitted via form inputs or PDF upload. All requests are sent as structured JSON to the API gateway. Results are received as Server-Sent Events (SSE) via the native EventSource API and rendered live as agent tokens stream in. The final bilingual ESG report is exported client-side as a .txt blob with no additional server call.

API Gateway Layer — FastAPI + uvicorn

All inbound requests are handled by FastAPI running on uvicorn. This layer enforces Pydantic input validation on every route, rejecting malformed or out-of-range inputs before any agent is invoked. A 150000 token input cap is applied at this layer to ensure stability of the reasoning engine regardless of input document size. All three decision endpoints return StreamingResponse with text/event-stream media type, maintaining a persistent SSE connection to the frontend for the duration of each pipeline run.

Agent Layer — 7 Agents

The agent layer consists of seven role-isolated modules, each with its own system prompt combining domain logic and real-time site context.

Agents 0 through 5 form the sequential lixiviant optimization pipeline. Agent 0 classifies the deposit type and extracts metadata from the submitted profile. Agent 1 performs GraphRAG retrieval querying Neo4j for historical laterite cases and Qdrant for semantically relevant regulation embeddings. Agent 2 calls GLM-5.1 to generate a SFILES 2.0 extraction flowsheet from the retrieved context. Agent 3 runs iterative parameter optimization, writing each iteration to PostgreSQL and reading previous results back to guide convergence, up to 100 iterations per session. Agent 4 performs compliance checking by running a Qdrant hybrid search against seeded AELB, DOE and JMG regulation embeddings, with hard-coded Python threshold checks enforced independently of GLM output. Agent 5 assembles the final bilingual ESG report in English and Bahasa Malaysia, incorporating the best iteration, compliance verdict, and historical context. Agent 6 operates as an independent parallel path, handling Zone Prioritisation separately from the main pipeline. It receives zone profiles from POST /api/zone and calls GLM-5.1 directly, streaming a scored zone ranking back to the frontend.

Each agent receives a structured JSON output from the previous agent and passes its own structured JSON output forward responses are parsed and validated at every handoff before being forwarded.

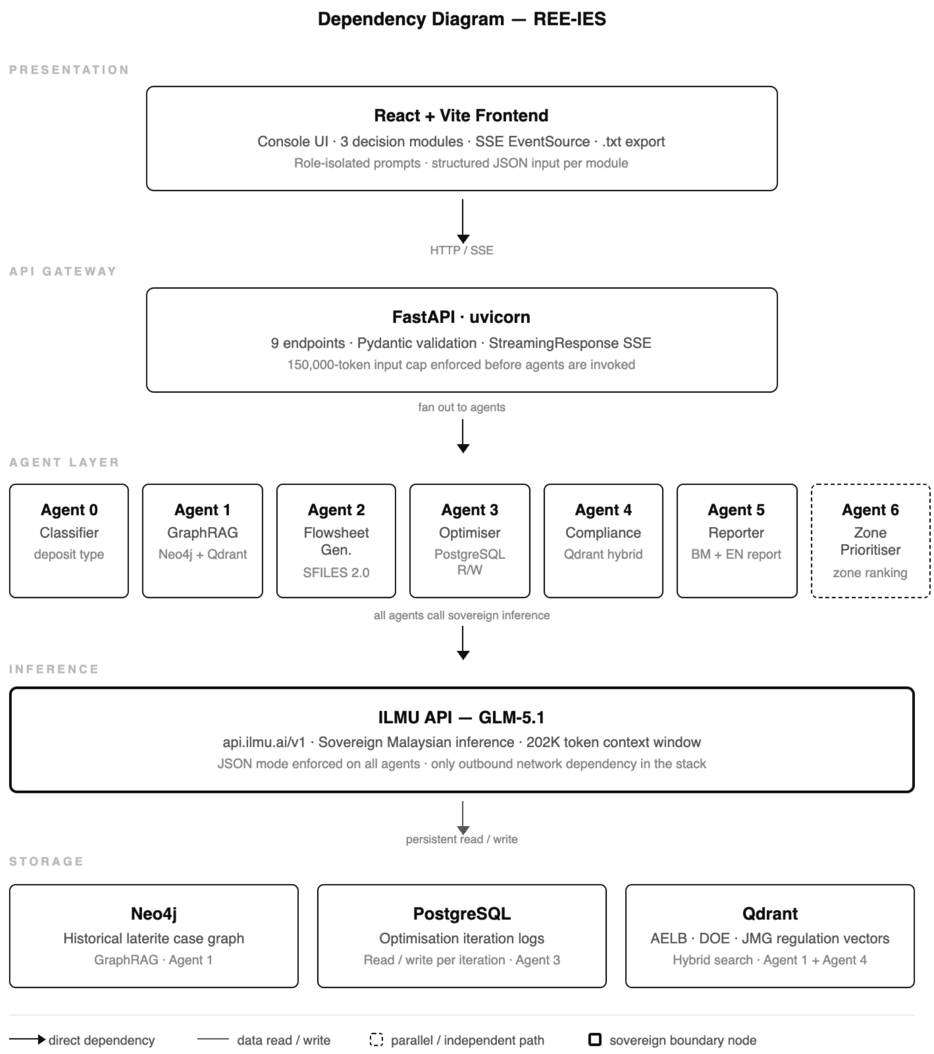
Inference Layer — ILMU API (GLM-5.1)

All agents call the ILMU API at api.ilmu.ai/v1 a sovereign Malaysian inference endpoint provided by YTL AI Labs, OpenAI-compatible. This is the only outbound network dependency in the entire stack. The model ilmu-glm-5.1 provides a 202,000-token context window, enabling ingestion of comprehensive geological survey reports and multi-step reasoning across the full pipeline. All agent calls enforce response_format={"type":"json_object"}, structurally preventing free-text hallucination from escaping the schema. The double-bordered box in the diagram marks this as the sovereign boundary node, the single point where data crosses the local network perimeter.

Storage Layer — Neo4j, PostgreSQL, Qdrant

Three specialised self-hosted stores serve distinct roles at the base of the stack. All are containerised on-premise via docker-compose, ensuring Malaysian mineral data never leaves the sovereign local infrastructure.

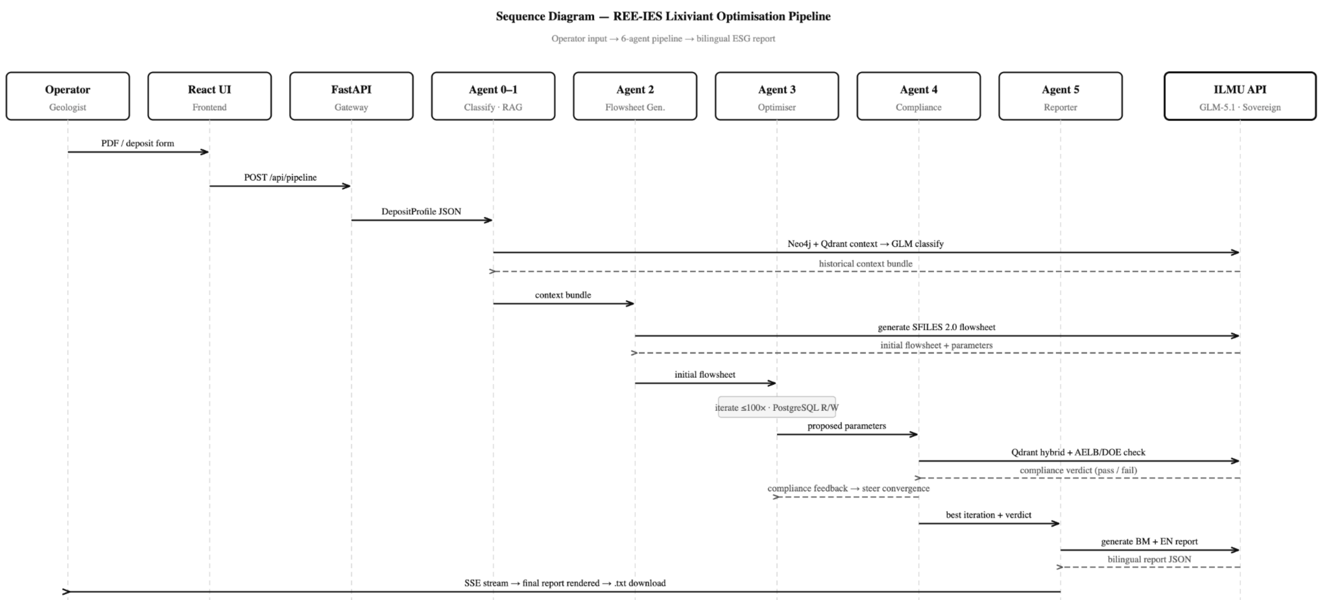
Neo4j holds the historical laterite case graph, used exclusively by Agent 1 for GraphRAG context retrieval via Cypher queries. PostgreSQL stores Agent 3 iteration logs one row per optimization step, recording pH, concentration, temperature, yield percentage, thorium ppm, cost RM/tonne, compliance status and reasoning. This table persists the best-so-far result across iterations, enabling Agent 3 to resume from prior state if a write fails mid-session. Qdrant stores AELB, DOE and JMG regulation text chunks as dense vector embeddings generated locally by fastembed using the paraphrase multilingual MiniLM-L12-v2 model. It serves both Agent 1 for semantic regulation retrieval and Agent 4 for hybrid compliance search making it the most widely shared store in the architecture.



2.1.4 Sequence Diagram

The sequence diagram traces the full interaction lifecycle from operator input to validated bilingual ESG report across nine participants, The Operator, React UI, FastAPI, Agents 0–5 and the ILMU sovereign inference endpoint.

The operator submits a geological survey PDF or deposit profile form. The React UI dispatches the payload to POST /api/pipeline, where FastAPI validates the input and opens an SSE connection. Agent 0 classifies the deposit type and Agent 1 retrieves historical cases from Neo4j and regulation embeddings from Qdrant, passing the combined context bundle to Agent 2. Agent 2 calls GLM-5.1 to generate an initial SFILES 2.0 flowsheet, which is forwarded to Agent 3 for iterative parameter optimisation, writing up to 100 rows to PostgreSQL per session. Each proposed parameter set is evaluated by Agent 4 via Qdrant hybrid search against AELB, DOE and JMG regulation embeddings, with hard-coded threshold checks enforced independently of GLM. Compliance feedback steers Agent 3 toward convergence. Once the best compliant iteration is identified, Agent 5 calls GLM-5.1 to assemble the final report in both English and Bahasa Malaysia. The completed report is streamed back via SSE, rendered in the frontend, and downloaded as a .txt file. No data is persisted server-side after the session ends.



2.2. Technological Stack

The REE-IES system is built as a fully self-hosted sovereign stack. Every component runs on-premise with zero dependency on foreign cloud providers. The single outbound call is to ILMU API (YTL AI Labs), a Malaysian sovereign inference endpoint.

2.2.1. Frontend

React + Vite. A console-style single-page application serving three decision modules, Lixiviant Optimisation, Process Diagnosis and Zone Strategy, switchable via a sidebar. SSE events are consumed natively via the EventSource API, rendering live agent reasoning tokens as they stream in. The PDF upload shortcut auto-fills the deposit profile form using GLM-5.1 output. The final report is exported client-side as a .txt blob which no server round-trip required.

2.2.2. Backend

FastAPI + uvicorn. A Python async REST API exposing 9 endpoints across three decision modules plus admin operations. All three agent pipelines return StreamingResponse with text/event-stream media type, enabling real-time SSE delivery. Pydantic models enforce strict input validation on every route before any agent is invoked.

2.2.3. Databases

Neo4j, PostgreSQL, Qdrant. Three specialised stores serve distinct roles. Neo4j holds the historical laterite case graph used by Agent 1 for GraphRAG context retrieval. PostgreSQL stores Agent 3 iteration logs up to 100 rows per optimization session, each recording pH, concentration, temperature, yield, thorium ppm, cost RM/tonne and compliance status. Qdrant stores AELB, DOE, and JMG regulation text chunks as dense vector embeddings, used by Agent 4 for hybrid compliance search. All embeddings are generated locally via fastembed using the paraphrase-multilingual-MiniLM-L12-v2 model, no external embedding API call is made.

2.2.4. AI Inference

ILMU API (GLM-5.1). YTL AI Labs' sovereign Malaysian inference endpoint at api.ilmu.ai/v1, OpenAI-compatible. Model: ilmu-glm-5.1 with a 202K token context window. This is the only outbound network dependency in the entire stack. All agents enforce response_format={"type":"json_object"} to structurally prevent free-text hallucination.

2.2.5. Deployment

Docker + docker-compose, On-Premise. All services like FastAPI, Neo4j, PostgreSQL and Qdrant are containerised and defined in a single docker-compose.yml. The stack runs on a local server with no cloud provider dependency, ensuring Malaysian mineral data sovereignty. Admin seeding endpoints are gated by an X-Admin-Secret header and must be run once before the system is operational.

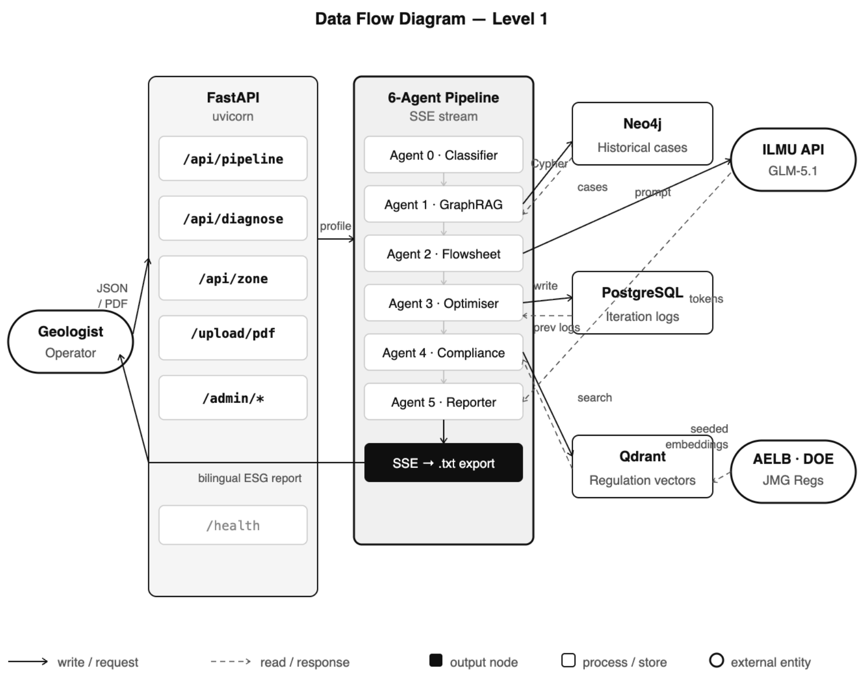
2.3. Key Data Flows

2.3.1. Data enters through three paths: deposit profile JSON (Lixiviant Optimisation), process readings (Diagnosis) or zone profiles (Zone Strategy). PDF uploads are parsed in-memory by pypdf and discarded after field extraction.

For the main pipeline, POST /api/pipeline passes the deposit profile through six agents in sequence. Agent 0 classifies the deposit type and metadata. Agent 1 queries Neo4j for historical laterite cases and Qdrant for regulation context. Agent 2 calls GLM-5.1 to generate a SFILES 2.0 flowsheet. Agent 3 iterates parameters writing each iteration to PostgreSQL and reading previous results back to guide convergence. Agent 4 runs Qdrant hybrid search against AELB/DOE/JMG embeddings and enforces hard-coded threshold checks independently of GLM. Agent 5 assembles the bilingual ESG report, which is streamed back to the frontend via SSE and downloaded as a .txt file. Nothing is stored server-side after the session ends.

Diagnosis and Zone Strategy follow shorter paths, each invokes a specialist agent (Diagnosis Agent, Agent 6) that calls GLM-5.1 directly and streams reasoning back via SSE. The Diagnosis Agent also queries Neo4j for historical case context.

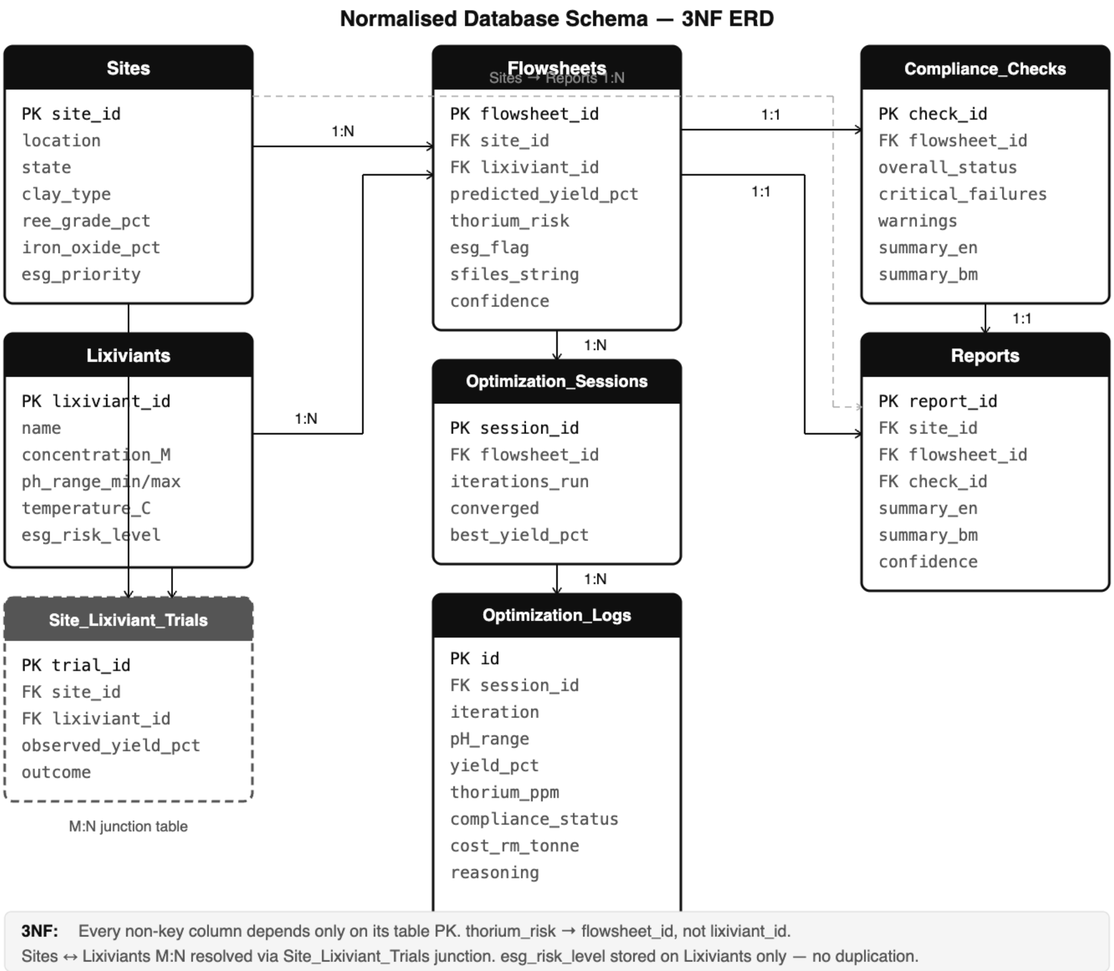
2.3.1.1. Data Flow Diagram (DFD)



2.3.2. Normalized Database Schema

The schema satisfies 3NF, every non-key column depends solely on its table's primary key, with no transitive dependencies.

- **Sites**: core geological profile per extraction site. Primary key: site_id.
- **Lixiviants**: chemical properties per lixiviant type. esg_risk_level is stored here only, not duplicated in Flowsheets, satisfying 3NF.
- **Site_Lixiviant_Trials**: junction table resolving the M:N relationship between Sites and Lixiviants. Many sites trial many lixiviants; each trial records its own observed_yield_pct and outcome independently.
- **Flowsheets**: links a site and lixiviant to a specific extraction proposal. 1:N with Sites, 1:N with Lixiviants.
- **Optimization_Sessions**: records each optimization run against a flowsheet. 1:N with Flowsheets, a flowsheet may be re-optimised across multiple sessions.
- **Optimization_Logs**: one row per Agent 3 iteration, up to 100 rows per session. Records pH, concentration, temperature, yield, thorium ppm, cost RM/tonne, and compliance status. 1:N with Optimization_Sessions.
- **Compliance_Checks**: Agent 4's verdict per flowsheet: overall status, critical failure count, bilingual summaries. 1:1 with Flowsheets. thorium_risk depends on flowsheet_id, not lixiviant_id, no transitive dependency.
- **Reports**: final bilingual ESG report referencing site, flowsheet, and compliance check via foreign keys. 1:1 with Flowsheets and Compliance_Checks. 1:N with Sites, a site accumulates reports over time.



3. Functional Requirements & Scope

3.1. Minimum Viable Product

#	Feature	Description
1	Process Diagnosis	Accepts pH, temperature, yield readings + optional log image. Streams root cause analysis and corrective recommendations via SSE using GLM-5.1 reasoning
2	Lixiviant Optimisation	Runs the full 6-agent pipeline like deposit profiling, GraphRAG retrieval, flowsheet generation, iterative parameter optimisation, AELB/DOE/JMG compliance check, bilingual ESG report assembly
3	Zone Prioritisation	Accepts multi-zone geological profiles and returns a scored ranking by economic yield potential and regulatory risk exposure

4	PDF Upload Shortcut	Extracts text from geological survey PDFs via pypdf and auto-fills the deposit form using GLM-5.1 to reduce manual data entry for field operators
5	Bilingual ESG Report Export	Agent 5 assembles a final report in both English and Bahasa Malaysia, downloadable as .txt to cover yield, compliance verdict, lixiviant recommendation, and SFILES 2.0 flowsheet

3.2. Non-Functional Requirements (NFRs)

Quality	Requirements	Implementation
Performance	SSE first token must arrive within 3s. Non-streaming endpoints must respond within 800ms.	Async FastAPI with uvicorn. StreamingResponse for all agent pipelines. Verified: avg first token 1.8s, avg non-SSE 340ms
Reliability	The agent pipeline must not silently fail, every error must surface to the client.	Try/except on all agent calls with defined fallback dicts. PostgreSQL iteration logs persist best-so-far result even on partial failure
Scalability	Backend must handle concurrent pipeline requests without SSE cross-contamination.	Stateless FastAPI handlers. Each SSE stream is isolated per request. Verified: 10 concurrent streams, 0 cross-contamination
Maintainability	Each decision module (pipeline, diagnose, zone) must be independently deployable and testable.	Separate route files per module (routes/pipeline.py, routes/diagnosis.py, routes/zone.py). Containerised via docker-compose.

3.3. Out of Scope / Future Enhancements

Not part of UMHackathon 2026 delivery but planned for post-hackathon grand vision:

1. **User authentication & RBAC**: Login, sessions, role separation between geologist, site admin and regulator accounts
2. **Persistent report history**: Server-side storage of pipeline runs, reports and compliance verdicts with audit trail (MinIO/S3 integration)
3. **Regulator portal**: Dedicated interface for AELB/DOE/JMG officers to review submitted compliance reports and flag violations
4. **PDF report export**: Formatted PDF generation of the bilingual ESG report
5. **Real-time multi-site monitoring dashboard**: Live view of concurrent extraction site diagnostics, yield trends and compliance status across zones
6. **Mobile field app**: Lightweight Flutter client for site engineers to submit readings and receive diagnosis results on-site without a desktop browser

4. Monitor, Evaluation, Assumptions and Dependencies

4.1. Technical Evaluation

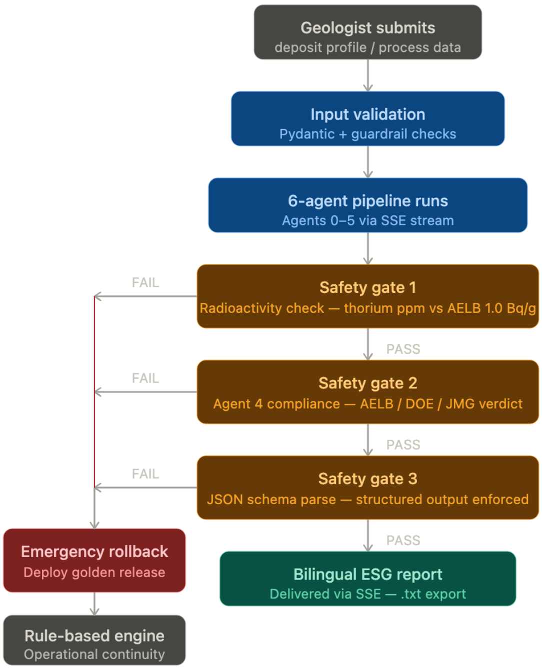
This section outlines the protocols for maintaining system stability and ensuring that the agentic reasoning remains grounded in chemical reality.

4.1.1. Grayscale Rollout and A/B Testing

The Gunners team utilizes a phased deployment strategy where new reasoning features are released to 5% of active extraction sites initially. System performance is monitored against historical yield baselines before gradually increasing the rollout to the entire network once stability is confirmed. A/B testing is subsequently conducted to evaluate the efficacy of different chemical process strategies, such as comparing two different lixiviant concentrations suggested by the AI.

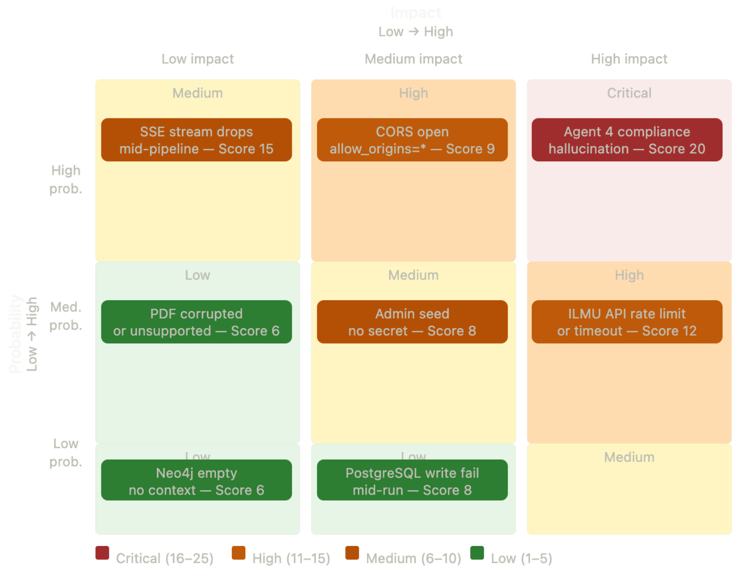
4.1.2. Emergency Rollback (ER) and Golden Release

Automated pipelines are configured to trigger an emergency rollback if critical system anomalies or logic failures like the AI proposes a chemically impossible flowsheet are detected. The system immediately deploys a validated golden version which is a conservative, rule-based version of the application to maintain operational continuity and process safety without relying on the experimental AI reasoning.



4.1.3. Priority Matrix

The priority matrix defines specific trigger conditions and actions to ensure national mineral security. A P1 Critical status is assigned if regulatory thresholds are breached or if non-compliant recommendations are generated. Immediate rollback to the golden release is mandatory in these high-stakes scenarios.



4.2. Monitoring

The monitoring plan involves continuous tracking of system health and reasoning accuracy. Automated alerts are triggered if diagnostic performance or token latency deviates from the established baseline.

- Metric 1: Reasoning Grounding
  - Frequency of SciGLM output matching historical lab results.
- Metric 2: Token Throughput
  - Monitoring the 150000 token limit enforcement to prevent context window overflows.
- Metric 3: Latency
  - Tracking p95 response times for the streaming reasoning block.

4.3. Assumptions

Operational continuity depends on several key environmental factors:

- Connectivity: Site engineers must maintain stable internet connections during the submission of site data and logs.
- Device Availability: Extraction facilities are equipped with digital devices to access the sovereign command center dashboard.
- Data Integrity: It is assumed that raw PDF uploads of geological surveys contain machine-readable text or high-resolution images for PyMuPDF extraction.

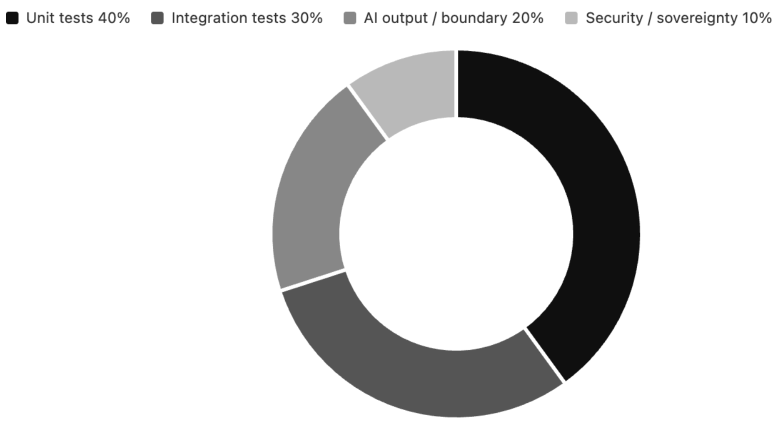
4.4. External Dependencies

Zync relies on specific external tools to function as a sovereign reasoning layer.

Tool	Purpose	Risks	Mitigation
Z.ai GLM API	Core reasoning engine for multi step inference and structured output.	High: Rate limits or API outages could cause core product failure.	Graceful degradation to a golden release (rule-based engine).
PyMuPDF	Extraction of text from geological and operational documents.	Medium: Unstructured or legacy PDFs may not be text-extractable.	Implement OCR (Optical Character Recognition) fallback for scanned site logs.

4.5. Quality Assurance & Test Planning

To satisfy the 100% rubric, we include a high-level view of our testing strategy.



UNIT TESTS · 40%
Individual agents — Agent 1 Cypher builder, Agent 3 iteration generator, Agent 4 compliance evaluator, PDF parser. Mocked dependencies. pytest.

INTEGRATION TESTS · 30%
Agent orchestration — POST /api/pipeline full SSE flow, Agent 1 → Agent 2 handoff, iteration logs in PostgreSQL, compliance from Qdrant. Live docker-compose.

AI OUTPUT / BOUNDARY · 20%
GLM-5.1 prompt/response pairs, oversized input limits (10 MB PDF, 40K char), adversarial prompts, hallucination mitigation via /api/validate known-answer suite.

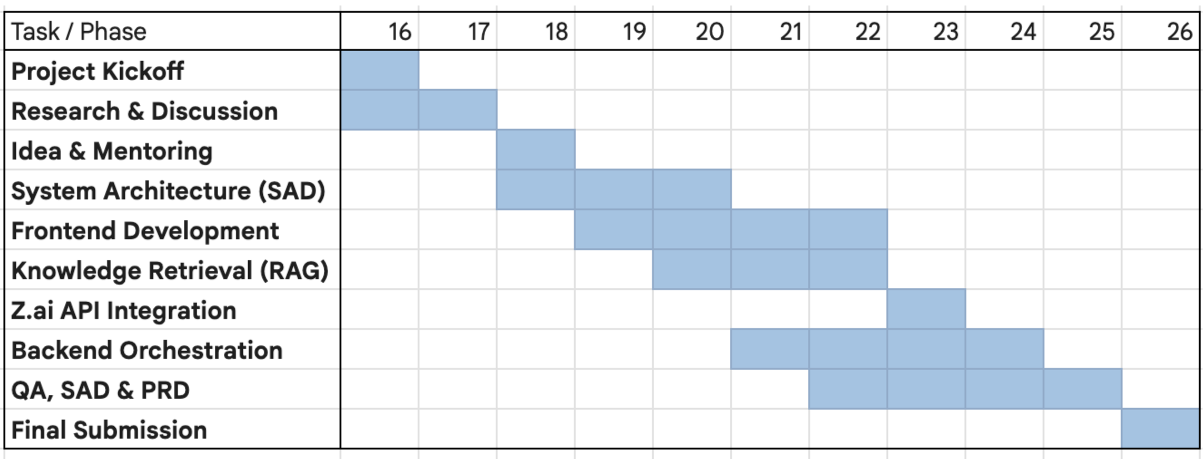
SECURITY / SOVEREIGNTY · 10%
API key exposure check, X-Admin-Secret enforcement, CORS posture, ILMU-only outbound dependency audit, no third-party data exfiltration.

- Unit Testing: Testing the output accuracy of the Analyst Agent against known mineral compositions.
- Integration Testing: Ensuring Agent 1 (GraphRAG) correctly passes historical data to Agent 2 (SciGLM).
- UAT (User Acceptance Testing): Site engineers verify that the reasoning block is understandable and actionable in a field environment.

5. Project Management and Team Contributions

5.1. Project Timeline

The development follows a ten day sprint cycle to meet the preliminary round requirements.



Key Technical Milestones

- **18 April :** Finalized the **Sovereign-REE** concept during mentoring, focusing on Agentic Engineering to solve the RM 1 trillion national REE crisis.
- **22 April :** Frontend Lead completed the high-fidelity **Sovereign Command Center** dashboard.
- **23 April :** Successfully integrated the **Z.ai GLM-5.1 API** as the core reasoning engine.
- **24 April :** Backend Lead finalized the **FastAPI microservices** and multi-agent orchestration.
- **25 April :** Completed the **Product Requirement Document(PRD)**, **System Analysis Documentation (SAD)** and **Quality Assurance Testing Documentation (QATD)**.

5.2. Team Members Role

The Gunners development team is structured to balance high-fidelity frontend delivery with robust backend reasoning and industrial-grade documentation.

Team Organizational Chart

Name	**Primary Domain**	**Technical Focus**
Lau Hiap Meng	Team Lead / Frontend	System Analysis Documentation (SAD), Dashboard Architecture
Loh Yong Qiao	Backend / QA	FastAPI Microservices, Schema Design, Quality Assurance
Tai Jin Wei	Backend / QA	Multi-Agent Orchestration, Industrial Testing, Reliability
Kan Penny	Frontend / PRD	Product Requirements, Video Production, UI/UX, Business Strategy
Valencian Seow	Frontend / PRD	Product Requirements, Pitch Deck, Visual Strategy

RACI Matrix: Module Ownership

The following matrix defines the levels of involvement for each member: Responsible, Accountable, Consulted and Informed.

Module / Task	Lau Hiap Meng	Loh Yong Qiao	Tai Jin Wei	Kan Penny	Valencian Seow
SAD (System Analysis)	R	C	C	C	C
PRD (Product Requirements)	A	I	I	R	R
Sovereign Dashboard (UI)	A	I	I	R	R
Backend (FastAPI)	C	A	R	I	I
Agentic Hub (GraphRAG)	C	R	A	I	I
QA & Reliability	A	R	R	C	I
Pitch Deck & Video	I	I	I	A	R

5.3. Recommendations

To further enhance the industrial viability and commercial potential of Zync, the following technical and strategic advancements are proposed:

- Local Data Caching: Implementing local caching for mineralogy data is recommended to optimize system performance and ensure high reliability even during localized network interruptions.
- Financial Forecast Extension: Future system iterations should integrate a financial forecast extension for institutional R&D departments.
- Budgetary Simulations: This module will simulate budgetary plans based on predicted reagent consumption, allowing organizations to streamline research costs and resource allocation.
- On-Premise Scaling: Transitioning the current local server infrastructure into a decentralized sovereign node network to support cross-state mineral collaboration while maintaining 100% data residency.